

EXPERIMENT-18

Implement Perceptron-based Iris Classification.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score, confusion_matrix

print("=" * 70)
print("      PERCEPTRON BASED IRIS CLASSIFICATION")
print("=" * 70)

uploaded = files.upload()
filename = list(uploaded.keys())[0]
data = pd.read_csv(filename)
print("\nIRIS DATASET")
print("-" * 70)
print(data.head(10))
X = data.iloc[:, 1:5]
y = data.iloc[:, 5]
encoder = LabelEncoder()
y = encoder.fit_transform(y)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
```

```

X_test = scaler.transform(X_test)
model = Perceptron(
    max_iter=1000,
    random_state=42
)
print("\nTraining Perceptron...")
model.fit(X_train, y_train)
prediction = model.predict(X_test)
print("\nTraining Completed.")
print("\nActual Values:")
print(list(y_test))
print("\nPredicted Values:")
print(list(prediction))
print("\nAccuracy:")
print(round(accuracy_score(y_test, prediction) * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, prediction))
sample = [[5.1, 3.5, 1.4, 0.2]]
sample_scaled = scaler.transform(sample)
result = model.predict(sample_scaled)
print("\nNew Flower:")
print(sample)
print("\nPredcted Flower:")
print(encoder.inverse_transform(result)[0])

```

Sample Input:

```

... =====
                        PERCEPTRON BASED IRIS CLASSIFICATION
=====
Choose Files Iris.csv
Iris.csv(text/csv) - 5107 bytes, last modified: 8/5/2026 - 100% done
Saving Iris.csv to Iris (2).csv

```

Sample Output:

```

IRIS DATASET
...
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0   1             5.1             3.5             1.4             0.2  Iris-setosa
1   2             4.9             3.0             1.4             0.2  Iris-setosa
2   3             4.7             3.2             1.3             0.2  Iris-setosa
3   4             4.6             3.1             1.5             0.2  Iris-setosa
4   5             5.0             3.6             1.4             0.2  Iris-setosa
5   6             5.4             3.9             1.7             0.4  Iris-setosa
6   7             4.6             3.4             1.4             0.3  Iris-setosa
7   8             5.0             3.4             1.5             0.2  Iris-setosa
8   9             4.4             2.9             1.4             0.2  Iris-setosa
9  10             4.9             3.1             1.5             0.1  Iris-setosa

Training Perceptron...

Training Completed.

Actual Values:
[np.int64(1), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(0), np.int64(1), np.int64(2), np.int64(1), np.int64(1), np.int64(2), np.int64(0), np.int64(0), np.i

Predicted Values:
[np.int64(1), np.int64(0), np.int64(2), np.int64(2), np.int64(2), np.int64(0), np.int64(0), np.int64(2), np.int64(1), np.int64(1), np.int64(2), np.int64(0), np.int64(0), np.i

Accuracy:
82.22 %

Confusion Matrix:
[[19  0  0]
 [ 1  5  7]
 [ 0  0 13]]

New Flower:
[[5.1, 3.5, 1.4, 0.2]]

Predicted Flower:
Iris-setosa

```